

Sean Phillips | Curriculum Vitae

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Short Biography

Sean Phillips is a Senior Mechanical Engineer and Technology Advisor of the Space Control Branch at the Air Force Research Laboratory in the Space Vehicles Directorate. He holds the title of Research Assistant Professor (LAT) at the University of New Mexico in Albuquerque, NM. He received his Ph.D in the Department of Computer Engineering at the University of California, Santa Cruz in 2018. He received his B.S. in Mechanical Engineering from the University of Arizona in 2011 and his M.S. in Mechanical Engineering from the same university, specializing in Dynamics and Controls, in 2013. In 2017, he received the Jack Baskin and Peggy Downes-Baskin Fellowship from the Baskin School of Engineering at the University of California, Santa Cruz. In 2024, he received the Air Force Research Laboratory Early Career Award.

Research Interests

Modeling, stability, robust control, observer design, and simulation of hybrid systems with applications to communication networks, multi-agent systems, power, aerospace/space, and biology.

Education

University of California, Santa Cruz <i>Ph.D. in Computer Engineering w/ Emphasis in Robotics and Control</i> Thesis Title: Coordination and Control of Networked Systems with Intermittent Communication	Santa Cruz CA 2014–2018
University of Arizona <i>M.S. in Mechanical Engineering</i> Thesis Title: Modeling and Analysis of Robust Stability for Spiking Neuron Models	Tucson AZ 2011–2013
University of Arizona <i>B.S. in Mechanical Engineering</i> Senior Capstone Project: ASME Human Powered Vehicle Competition	Tucson AZ 2006–2011

Professional Experience

Technology Advisor, Space Control Branch <i>Air Force Research Laboratory – Space Vehicles/Warfare Directorate</i> Leads research and development of Space Warfighting technologies to meet the needs of the US Space Force and US SPACECOM. Guides tactical and strategic planning for the technology areas of the branch including Autonomy, Cyber and XGEO Space Situational Awareness of a \$150M portfolio including multiple flight experiments. Ensuring that the technical progress is of the highest quality with aim to transition technologies to operational and acquisition offices.	2024–present Albuquerque, NM
Senior Mechanical Engineer (DR-III) <i>Air Force Research Laboratory – Space Vehicles/Warfare Directorate</i> Autonomous Systems and Controls Engineer for the Space Control Branch who is personally responsible for applying advanced principles of dynamics and controls for distributed multi-agent autonomous satellite systems through the applications of a variety of techniques. Responsible for the development of verification and validation of autonomous and guidance navigation and control algorithms for satellite systems. Founder and manager of the Local Intelligent Networked Clustered Satellites (LINCS) Lab. Actively managing a fast-paced \$40M portfolio spanning basic to advanced R&D across academic, industrial, other national governmental, and international organizations to research and develop next-generation technologies to meet US Space Force, US SPACECOM and other government agency needs.	2023–present Albuquerque, NM

Research Assistant Professor (LAT) <i>University of New Mexico – Electrical and Computer Engineering Department</i> Teaches 1 course annually and actively mentors students in the ECE and ME departments.	2019–present Albuquerque, NM
Deputy Technology Advisor, Space Control Branch <i>Air Force Research Laboratory – Space Vehicles Directorate</i> Helped assure the highest R&D standards in the technical development of the AFRL Space Control Branch to meet the needs of the US Space Force, US SPACECOM, and allied partners.	2019–2024 Albuquerque, NM
Research Mechanical Engineer (DR-II) <i>Air Force Research Laboratory – Space Vehicles Directorate</i>	2018–2023 Albuquerque, NM
Graduate Student Researcher <i>Hybrid Systems Lab, University of Santa Cruz</i>	2014–2018 Santa Cruz, CA
Graduate Teaching Assistant <i>Computer Engineering Department, University of Santa Cruz</i>	2015–2016 Santa Cruz, CA
Graduate Research Assistant <i>Hybrid Dynamics and Controls Lab, University of Arizona</i>	2011–2014 Tucson, AZ
Graduate Teaching Assistant <i>Aerospace and Mechanical Engineering Department, University of Arizona</i>	2012–2011 Tucson, AZ
Space Scholar Graduate Intern <i>Air Force Research Lab, Kirtland Air Force Base</i>	Summers in 2011,'12,'17 Albuquerque, NM
NASA Space Grant Internship <i>Lunar and Planetary Laboratory, University of Arizona</i>	2010–2011 Tucson, AZ
Undergraduate Research Assistant <i>Hybrid Dynamics and Controls Lab, University of Arizona</i>	2009–2011 Tucson, AZ

Honors

2nd Quarter CAT-III Award <i>AFRL/RVS Spacecraft Technologies Division</i>	Kirtland AFB, NM 2026
STEM Award <i>AFRL/RVS Spacecraft Technologies Division</i>	Kirtland AFB, NM 2026
Director's Cup Senior Individual Award <i>AFRL/RV Space Vehicles Directorate</i>	Kirtland AFB, NM 2025
Best Paper Award <i>NATO AVT-417 Enabling Resilient and Adaptive HSA for Defence and Security Missions</i>	Riga, LAT 2025
Early Career Award <i>Air Force Research Laboratory</i>	Wright Patterson AFB, OH 2024
Best Team Paper Award <i>AFRL/RV Space Vehicles Directorate</i>	Kirtland AFB, NM 2024
International Achievement Award <i>AFRL/RVSW Space Control Branch</i>	Kirtland AFB, NM 2024
Leadership Award <i>AFRL/RVSW Space Control Branch</i>	Kirtland AFB, NM 2024
Notable Achievement Award <i>AFRL/RV Space Control Branch</i>	Kirtland AFB, NM 2023
Best Individual Paper Award <i>AFRL/RVS Space Technology Division</i>	Kirtland AFB, NM 2023

Mentorship Award <i>AFRL/RV Space Control Branch</i>	Kirtland AFB, NM 2021
General Raymond A. Gilbert Junior Scientific Achievement Award <i>AFRL/RV Spacecraft Technology Division</i>	Kirtland AFB, NM 2020
Team Notable Achievement Award <i>AFRL/RV Spacecraft Technology Division</i>	Kirtland AFB, NM 2019
Jack Baskin and Peggy Downes-Baskin Fellowship <i>UCSC Jack Baskin School of Engineering</i>	Santa Cruz, CA 2016-2017
Computer Engineering Summer 2016 Fellowship <i>UCSC Computer Engineering Dept.</i>	Santa Cruz, CA 2016
Best Presentation in Session Award <i>IEEE American Control Conference</i>	Portland, OR 2014
Best Presentation in Session Award <i>IEEE American Control Conference</i>	Washington, DC 2013
Aerospace and Mechanical Eng. Department Graduate School Fellowship <i>University of Arizona</i>	Tucson, AZ 2012-2011
3rd Place in Design <i>ASME Human Powered Vehicle East Competition</i>	Indianapolis, IN 2011
NASA Space Grant Fellowship <i>Lunar and Planetary Sciences Dept., University of Arizona</i>	Tucson, AZ 2011-2010
Honors College Undergraduate Research Grant <i>University of Arizona</i>	Tucson, AZ 2010

Teaching Experience

2026: Instructor: Spacecraft Attitude Dynamics and Control (ME 596)	<i>University of New Mexico, Albuquerque, New Mexico</i>
2026: Lecture: Workshop on Data Driven Methods for Space Resiliency	<i>American Control Conference, New Orleans, LA</i>
2025: Instructor: Spacecraft Attitude Dynamics and Control (ME 596)	<i>University of New Mexico, Albuquerque, New Mexico</i>
2024: Instructor: Spacecraft Attitude Dynamics and Control (ME 596)	<i>University of New Mexico, Albuquerque, New Mexico</i>
2023: Guest Lecturer: Nonlinear and Adaptive Control (ECE 514)	<i>University of New Mexico, Albuquerque, New Mexico</i>
2020: Instructor: Undergraduate Course: Design of Feedback Control Systems	<i>University of New Mexico, Albuquerque, New Mexico</i>
2019: Guest Lecturer: Guidance, Navigation and Control of Spacecraft (ECE 595)	<i>University of New Mexico, Albuquerque, New Mexico</i>
2019: Lecturer: Workshop on Hybrid Model Predictive Control	<i>Conf. on Decision and Control, Nice France</i>
2019: Instructor, Undergraduate Course: Design of Feedback Control Systems	<i>University of New Mexico, Albuquerque, New Mexico</i>
2016: Lecturer: Workshop on Hybrid Control Systems	<i>Conf. on Decision and Control, Las Vegas NV</i>
2016: Substitute Lecturer: Introduction to Feedback Control (CMPE 8)	<i>U.C., Santa Cruz, CA</i>
2016: Grad TA: Intro. to Probability and Statistics (CMPE 107)	<i>U.C., Santa Cruz, CA</i>
2016: Lecturer: Introduction to Feedback Control for Middle School Girls in Engineering Society	<i>U.C., Santa Cruz, CA</i>
2015: Grad TA: Robot Automation: Intro. to Feedback Control (CMPE 8)	<i>U.C., Santa Cruz, CA</i>

2013: Lecturer: Introduction to Feedback Control for High School Teachers in MESA	University of Arizona
2011 – 2012: Grad TA: Instrumentation Lab (AME 200)	University of Arizona
2011: Grad TA: Engineering Analysis (AME 201)	University of Arizona

Professional Memberships

IEEE: Institute of Electrical and Electronics Engineers, Member, 2011 - present
IEEE-CSS: Control Systems Society, Member, 2014 - present
IEEE-CSS-HSTC: Hybrid Systems Technical Committee, Member, 2016 - present
IEEE-CSS-AC: Aerospace Control Technical Committee, Member, 2024 - present
AIAA: American Institute of Aeronautics and Astronautics, Member, 2020 - present
AIAA-ISTC: Intelligent Systems Technical Committee, Member, 2024 - present
AAS: American Astronomical Society, Member, 2021 - present

Theses

Doctorate of Philosophy in Computer Engineering.....

Title: *Coordination and Control of Networked Systems with Intermittent Communication*

Advisor: Dr. Ricardo G. Sanfelice

Abstract: Communication networks characterize many modern real-world processes, ranging from the internet and social media networks, to communication networks and power distribution networks, to interconnected neuron and biological models. For such networks, the communication coupling between agents may not be continuous, more specifically, the information transmitted between agents are naturally sporadically impulsive or intermittent. In this thesis, robust coordination algorithms are studied for the case when the agents have linear and hybrid dynamics when information coupling between the agents in the network are intermittent. Namely, coordination in terms of synchronization and desynchronization for a variety of systems are considered. The robust exponential stability of the set characterizing synchronization in continuous linear time-invariant systems when information from neighbors is received impulsively at isolated time instances. For each case, we model the interconnected agents as a hybrid system and use tools for robust set stability of hybrid systems to analyze the dynamic behavior. Applications to the study of dynamical behavior of spiking neurons modeled as a form of interconnected impulse-coupled oscillators and to frequency rendezvous for communication systems are given.

Master of Science in Mechanical Engineering.....

Title: *Modeling and Analysis of Robust Stability for Spiking Neuron Models*

Advisor: Dr. Ricardo G. Sanfelice

Abstract: A generalized hybrid systems framework for modeling and analysis of robust stability in spiking neurons is proposed. This framework is developed for a population of N-interconnected neurons to study the behavior of several excitatory and inhibitory neuron phase response models and to study the N-interconnected inhibiting integrate-and-fire neuron model. Using Lyapunov stability tools for hybrid systems, the stability properties for each case of the neuron models considered are established. Due to the regularity properties of the data of the hybrid models, the asserted stability properties are robust to small perturbations. For the integrate and fire model, analysis of several different perturbation types is given. Furthermore, an external stimuli is introduced to achieve a global asymptotic stability property for the 2-neuron simplified Hodgkin and Huxley neuron model. Numerical simulations are included to illustrate the results.

Bachelor of Science in Mechanical Engineering.....

Title: *Senior Design: ASME Human Powered Vehicle Competition*

Advisor: Dr. Cholik Chan and Mr. Joseph Hartley

Abstract: The national student chapters of the American Society of Mechanical Engineering has a competition each year to design, build and compete a customized human powered vehicle. In the academic year of 2010-2011, the University of Arizona supported its first competition in 20 years. The team designed a novel recumbent bicycle first in chromoly steel and redesigned in aluminum. The team won 3rd in Design at the ASME HPVC in Indianapolis, Indiana in April 2011

Publications

Book Chapters.....

[2] F. Zegers, **S. Phillips** and W. Dixon. Event-Driven Distributed Cyber-Physical Systems. In: Y. Wan, K. Vamvoudakis, Y. Chen, F. Lewis (eds) Smarter Cyber-Physical Systems: Enabling Methodologies and Applications. Springer. 2026

[1] **S. Phillips**, C. Petersen, and R. Fierro. Robust, Resilient, and Energy-Efficient Satellite Formation Control. In: Blondin, M.J., Fernandes Trovão, J.P., Chaoui, H., Pardalos, P.M. (eds) Intelligent Control and Smart Energy Management. Springer Optimization and Its Applications, vol 181. Springer, Cham. 2022.

Journal Articles.....

[20] C. W. Hays and **S. A. Phillips**. Necessary and Sufficient Conditions for State Omniscience of Linear Time-Invariant Distributed Estimators. IEEE Transactions on Control of Network Systems (under development)

[19] V. Sivaramakrishnan, R. Devonport, J.Y. Pilipovsky, K.C. Kalagarlad, P. Tsiotras, M.M.K. Oshib, and **S.A. Phillips**. SAVER-Tubes: SAMpling-Based VERification of Black-Box Systems With Stochastic Reach Tubes. Nonlinear Analysis: Hybrid Systems (under review)

[18] Kealy A. Murphy, M. R. Mercier, David H. Curtis, and **S. A. Phillips**. Rapid Passive Safety Analysis Applied to RPO Mission Planning and Runtime Assurance AIAA Journal of Guidance Control and Dynamics (under review)

[17] O. J. R. Chuy, M. T. Hale, V. Sivaramakrishnan, **S. Phillips**, and R. G Sanfelice. Autonomous Satellite Rendezvous via Hybrid Feedback Optimization. Nonlinear Analysis: Hybrid Systems Special Issue on Applications of Hybrid Systems (under review)

[16] G. Behrendt, M. Hale, A. Soderlund, **S. Phillips**, and E. Kain. Time-Constrained Model Predictive Control for Autonomous Satellite Rendezvous, Proximity Operations, and Docking. AIAA Journal of Guidance Control and Dynamics (under review)

[15] F. M. Zegers and, **S. Phillips**. Consensus over clustered networks using intermittent and asynchronous output feedback. Nonlinear Analysis: Hybrid Systems vol. 59, 2026

[14] D. Grange, Y. Sun, C. Weistuch, R. Elkin, **S. Phillips**, and J. O. Deasy. Curvature on Graphs With Negative Edge Weights. IEEE Transactions on Network Science and Engineering, vol. 13, pp. 9205-9222, 2026.

[13] C. F. Nino, O. S. Patil, C. D. Petersen, **S. Phillips**, and W. E. Dixon. Collaborative Spacecraft Servicing Under Partial Feedback Using Lyapunov-Based Deep Neural Networks. Journal of Astronautical Sciences. vol 72, 2025.

[12] F. M. Zegers, **S. Phillips**, and G. P. Hicks. Timer-Based Coverage Control for Mobile Sensors. IEEE Transactions on Control of Network Systems, vol. 12, no. 2, pp. 1600-1612, 2025.

[11] C. W. Hays, T. Henderson, K. Miller, **S. Phillips**, and A. Soderlund. Angles-Only Cooperative Local Catalog Maintenance of Close-Proximity Satellite Systems Journal of Guidance, Control, and Dynamics Vol 47 No. 12, pp. 2573-2586, 2024

[10] **S. Phillips** and R. G. Sanfelice. Point-wise Exponential Stability of State Consensus with Intermittent Communication, *IEEE Transactions on Automatic Control*, vol. 69, no. 11, pp. 7540-7555, 2024.

[9] M. Abate, M. Mote, M. Dor, C. Klett, **S. Phillips**, K. Lang, P. Tsiotras, E. Feron, and S. Coogan. Run Time Assurance for Spacecraft Attitude Control Under Nondeterministic Assumptions. IEEE Transactions on Control Systems Technology. vol. 32, no. 3, pp. 862-873, 2024

- [8] J. Aurand, S. Cutlip, H. Lei, K. Lang, and **S. Phillips**. “Deep Q-Learning for Decentralized Multi-Agent Inspection of a Tumbling Target,” *AIAA Journal of Spacecraft and Robotics*, Vol. 61, No. 2, 2024.
- [7] D. Grange, M. Al-Jarrah, R. Baptista, A. Taghvaei, T. T. Georgiou, **S. Phillips**, and A. Tannenbaum, “Computational Optimal Transport and Filtering on Riemannian Manifolds,” *IEEE Control Systems Letters*, Vol. 7, pp. 3495-3500, 2023
- [6] M. Hibbard, M. Cubuktepe, M. Shubert, K. Lang, U. Topcu, **S. Phillips**. Trajectory Synthesis for the Coordinated Inspection of a Spacecraft with Provable Safety Guarantees. *AIAA Journal of Guidance Control and Dynamics*. Vol. 46, Issue 12, pp. 2245-2264, 2023
- [5] A. Soderlund and **S. Phillips**. Hybrid Systems Approach to Autonomous Rendezvous and Docking of an Underactuated Satellite. *AIAA Journal of Guidance, Control, and Dynamics*. Vol. 46, Issue 10, pp. 1901-1918, 2023. **2023 AFRL/RV Publication of the Year Team Award**
- [4] **S. Phillips** and R. G. Sanfelice. Robust Distributed Synchronization of Networked Linear Systems with Intermittent Information, *Automatica*, 105:323-333, 2019.
- [3] Y. Li, **S. Phillips**, and R. G. Sanfelice. Robust Distributed Estimation for Linear Systems under Intermittent Information, *IEEE Transactions on Automatic Control*, 63(4):973–988, 2018.
- [2] Y. Li, **S. Phillips**, and R. G. Sanfelice. Basic Properties and Characterization of Incremental Stability Prioritizing Flow Time for a Class of Hybrid Systems, *Systems and Control Letters*, 90:7–15, 2016.
- [1] **S. Phillips**, and R. G. Sanfelice. Robust asymptotic stability of desynchronization in impulse-coupled oscillators., *IEEE Transactions on Control of Networked Systems*, 3(2):127–136, 2016.

Conference Proceedings.....

- [76] S. E. Clees, **S. Phillips** and C. Petersen. Information Aware Model Predictive Control for Satellite Inspection. In Proceedings of the AAS Astrodynamics Specialists Conference, Whistler, CAN, 2026.
- [75] D. Doyle, E. Glass, A. Yezzi and **S. Phillips**. Beyond Point Features: Direct Boundary Fitting and Pose-Gradient Methods for Space Object Imaging. In Proceedings of the AAS Astrodynamics Specialists Conference, Whistler, CAN, 2026.
- [74] D. Reid, A. Releford, A. Andy, S.T. Smith, M. Alfred, and **S. Phillips**. A Decentralized Framework for Resource-Constrained Task Redistribution. In Proceedings of the AAAI Symposium Series 8 (1), 97-104, Burlingame, CA, 2026.
- [73] G.Newcomb, L. Vance, **S. Phillips** and J. Thangavelautham. Utilizing Blender to Duplicate Radiometrically Correct Satellite Imagery with Realistic Rotation. In Proceedings of the 47th Rocky Mountain AAS GN&C Conference, Breckenridge, CO, 2026.
- [72] Alexis Munguía, Leonard Vance, **S. Phillips** and J. Thangavelautham. Recovery of Cooperative Satellite Orientation and Range from Semi-Resolved Passive Color Imagery. In Proceedings of the 47th Rocky Mountain AAS GN&C Conference, Breckenridge, CO, 2026.
- [71] J. Ludwig, D. Touhy, H Schaub, A Cheval, **S. Phillips** and K. Miller. An Architecture for Trusted Long-Duration Satellite Autonomy. In Proceedings of AIAA SciTech 2026 Forum, Orlando, FL 2026.
- [71] **S. Phillips** and C. W. Hays. On Autonomous Satellite Constellations Towards a Future Hybrid Space Architecture. In Proceedings of NATO AVT-417 Research Specialists Meeting on Platform Implications for Hybrid Space Architectures for NATO, Riga, LAT 2026. **Best Paper Award for NATO AVT-417**
- [70] A. Reliford, D. Reid, S.T. Smith, A. Andy, M. Alfred, and **S. Phillips**. Autonomous Task Rescheduling in a Heterogeneous LEO Satellite Constellation Using Reinforcement Learning. In Proceedings of AIAA SciTech 2026 Forum, Orlando, FL 2026.
- [69] I. Paranjape, J. Murray-Krezan, **S. Phillips**, S. Chakravorty, and I. Hussein. Gaussian Mixture Model Fusion Across Decentralized Sensor Networks. In Proceedings of 2025 IEEE International Conference on Multisensor Fusion and Integration for Intelligent Systems (MFI), College Station, TX 2025.

- [68] D. Baker, J. Wojcik, and **S. Phillips**. Autonomy at Levels for Spacecraft. In Proceedings of 2025 AAS/AIAA Astrodynamics Specialist Conference, Boston, MA 2025.
- [67] D. Baker, J. Wojcik, and **S. Phillips**. Autonomy at Levels for Spacecraft. In Proceedings of 2025 AAS/AIAA Space Flight Mechanics Meeting, Kaua'i, HI 2025.
- [66] J. Murray-Krezan, J. Baker, S. Bonbrest, H. Borowski, I. Hussein and **S. Phillips**. Experimental Demonstrations of Trusted Distributed Autonomy Applications. In Proceedings of 46th Rocky Mountain AAS GN&C Conference. Breckenridge, CO. 2025.
- [65] Z. S. Lippay, **S. Phillips** and A. Maghareh. Multiagent Satellite Autonomy: A Benchmark Evaluation Framework. In Proceedings of 46th Rocky Mountain AAS GN&C Conference. Breckenridge, CO. 2025.
- [64] **S. Phillips**, J. Aurand and Z. S. Lippay. CLOSE-PROXIMITY SATELLITE OPERATIONS THROUGH DEEP REINFORCEMENT LEARNING AND TERRESTRIAL TESTING ENVIRONMENTS. In Proceedings of 46th Rocky Mountain AAS GN&C Conference. Breckenridge, CO. 2025.
- [63] D. Baker and **S. Phillips**. Levels of Spacecraft Autonomy. In Proceedings of 46th Rocky Mountain AAS GN&C Conference. Breckenridge, CO. 2025.
- [62] J. Aurand, C. Pang, S. Mokhtar, H. Lei, S.C. Cutlip, and **S. Phillips**. Assessing autonomous inspection regimes: Active versus passive satellite inspection. In Proceedings of the 2025 AIAA SciTech Forum, Orlando, FL 2025
- [61] H. Lei, Z. Lippay, A. Zaman, J. Aurand, A. Maghareh, and **S. Phillips**. Stability Analysis of Deep Reinforcement Learning for Multi-Agent Inspection in a Terrestrial Testbed. In Proceedings of the 2025 AIAA SciTech Forum, Orlando, FL 2025
- [60] E. Jose, R. Batta, M. Sudit, D. Sudit, J. L. Crassidis, C. Nebelecky, **S. Phillips**, and A. Soderlund. Optimizing Potential Information Gain From Satellites Over Space and Time for Heterogeneous Information Fusion. In Proceedings of the 2025 AIAA SciTech Forum, Orlando, FL 2025
- [59] K. Dunlap, N.P. Hamilton, Z. Lippay, M. Shubert, **S. Phillips**, and K.L. Hobbs. Demonstrating reinforcement learning and run time assurance for spacecraft inspection using unmanned aerial vehicles. In Proceedings of the 2025 AIAA SciTech Forum, Orlando, FL 2025
- [58] D. A. Copp and **S. Phillips**. Robust Hybrid Wide-Area Damping Control for Power Systems with Communication Errors. In Proceedings of the 2024 American Control Conference. Toronto, CAN 2024.
- [57] K. Miller, **S. Phillips** and S. Mitra. Guaranteed safe satellite guidance and navigation using reachability-based switching controllers. In Proceedings of the 2024 American Control Conference. Toronto, CAN 2024.
- [56] C.W. Hays, **S. Phillips** and T. Henderson. A necessary and sufficient condition for state omniscience of linear time-invariant distributed estimators. In Proceedings of the 2024 American Control Conference. Toronto, CAN 2024.
- [55] D. Grange, R. Baptisa, A. Taghvaei, A. Tannenbaum, **S. Phillips**. Distributed nonlinear filtering using triangular transport maps. In Proceedings of the 2024 American Control Conference. Toronto, CAN 2024.
- [54] J. Murray-Krezan, I. I. Hussein, H Borowski, J. Baker, C. Elliott, and **S. Phillips**. Simulation-Based Results of the Trusted Distributed Autonomy Demonstration Experiment. In Proceedings of the 2024 IEEE Aerospace Conference. Big Sky, MT. 2024.
- [53] J. Aurand, Z. Lippay, S. Cutlip, H. Lei, M. Shubert, and **S. Phillips**. Stability Analysis of Deep Reinforcement Learning for Multi-Agent Inspection in a Terrestrial Testbed. In Proceedings of 46th Rocky Mountain AAS GN&C Conference. Breckenridge, CO. 2024.
- [52] I. Paranjape, S. Chakravorty, **S. Phillips**, I. I. Hussein, J. Murray-Krezan, H Borowski, J. Baker. Multi-Sensor Tasking Strategies for Situational Awareness. In Proceedings of 46th Rocky Mountain AAS GN&C Conference. Breckenridge, CO. 2024.

- [51] H Borowski, I. I. Hussein, **S. Phillips**, J. Murray-Krezan, J. Baker, and C. Elliott. Application of Trusted Distributed On-board Autonomous Resource Management. In Proceedings of 46th Rocky Mountain AAS GN&C Conference. Breckenridge, CO. 2024.
- [50] C. Hays, T. Henderson, K. Miller, **S. Phillips** and A. Soderlund. State Omniscience for Cooperative Local Catalog Maintenance of Close Proximity Satellite Systems. In Proceedings of the 2024 AIAA SciTech Forum, Orlando, FL 2024
- [49] M. Stephenson, L. Mantovani, **S. Phillips**, and H. Schaub. Using Enhanced Simulation Environments to Improve Reinforcement Learning for Long-Duration Satellite Autonomy. In Proceedings of the 2024 AIAA SciTech Forum, Orlando, FL 2024
- [48] SC Cutlip, J Aurand, K Lang, **S. Phillips**. Multi-Agent Passive Inspection Coverage of an Unknown Torque-Free Rigid Body Using Monte Carlo Analysis and Quaternion Measurements. In Proceedings of the 2024 AIAA SciTech Forum, Orlando, FL 2024
- [47] M.J. Park, D. Markovic, M. Mesbahi, and **S. Phillips**. Extended Abstract for Collaborative Mission Autonomy for Resilient Highly Integrated Networked Operations (CMA-RHINO). In Proceedings of the 2024 AIAA SciTech Forum, Orlando, FL 2024
- [46] **S. Phillips**, Z. Lippay, D. Baker, A.A. Soderlund, and M. Shubert. Emulation of close-proximity spacecraft dynamics in terrestrial environments using unmanned aerial vehicles. In Proceedings of the 2024 AIAA SciTech Forum, Orlando, FL 2024
- [45] R.M. McCarthy, T. Thomas, C. Danielson, **S. Phillips**, and R. Fierro. Control for an omnidirectional multi-rotor UAV for space applications. In Proceedings of the 2024 AIAA SciTech Forum, Orlando, FL 2024
- [44] G Behrendt, A. Soderlund, M. Hale, and **S. Phillips**. Autonomous Satellite Rendezvous and Proximity Operations with Time-Constrained Sub-Optimal Model Predictive Control. In Proceedings of the 2023 IFAC World Congress. Yokohama, JAP 2023.
- [43] D. Grange, R. Sandhu, A.A. Soderlund, and **S. Phillips**. Consensus on region-based pose estimation for satellites. In Proceedings of the 2023 IEEE/ION Position, Location and Navigation Symposium (PLANS), pgs. 1322-1329. Monterey, CA 2023.
- [42] C. Petersen, R. J. Caverly, **S. Phillips** and A. Weiss. Safe and Constrained Rendezvous, Proximity Operations and Docking. In Proceedings of the 2023 American Control Conference. San Diego, CA 2023.
- [41] K. Miller, J. M. Brewer, A. Soderlund, and **S. Phillips**. Sensor Safety and Multi-Objective Satellite Control under Nonlinear Dynamics. In Proceedings of the 2023 American Control Conference. San Diego, CA 2023.
- [40] C. W. Hays, A. Soderlund, **S. Phillips**, and T. Henderson. Ubiquitous Controllability of Single Input Linear Time-Invariant Systems. In Proceedings of the 2023 American Control Conference. San Diego, CA 2023.
- [39] K. Hobbs, **S. Phillips**, J. Lyons, M. S. Feather, M. Simon, K. Costello, B. P. Bycroft, M. Harter, R. Fitz. Towards Space Trusted Autonomy Readiness Levels. In Proceedings of the 2023 IEEE Aerospace Conference. Big Sky, MT. 2023.
- [38] V. Bajenaru, H. Schaub, **S. Philips**, J. Ramirez, and A. Herrmann. Trustworthy Reinforcement Learning for Decentralized Control of Satellites. In Proceedings of 45th Rocky Mountain AAS GN&C Conference. Breckenridge, CO. 2023.
- [37] C. Hays, **S. Philips**, K. Miller, A. Soderlund and T. Henderson. Local Catalog Maintenance of Close Proximity Satellite Systems on Closed Natural Motion Trajectories. In Proceedings of 45th Rocky Mountain AAS GN&C Conference. Breckenridge, CO. 2023.
- [36] V. Chen, **S. Philips**, and D. Copp. Planning Autonomous Spacecraft Rendezvous and Docking Trajectories via Reinforcement Learning. In Proceedings of 45th Rocky Mountain AAS GN&C Conference. Breckenridge, CO. 2023.

- [35] J. Aurand, S. Cutlip, H. Lei, K. Lang, and **S. Philips**. Exposure-Based Multi-Agent Inspection of a Tumbling Target Using Deep Reinforcement Learning. In Proceedings of 45th Rocky Mountain AAS GN&C Conference. Breckenridge, CO. 2023.
- [34] C. Staudt, C. Zagaris, and **S. Philips**. Sliding Mode Control of Satellite Formations Under Imperfect Communications via Network Trust. In Proceedings of 45th Rocky Mountain AAS GN&C Conference. Breckenridge, CO. 2023.
- [33] A. Dang, D. Copp, and **S. Philips**. Benchmarking State Estimation Algorithms for Output Feedback Model Predictive Control of Spacecraft Autonomous Rendezvous and Docking. In Proceedings of 45th Rocky Mountain AAS GN&C Conference. Breckenridge, CO. 2023.
- [32] A. C. Nichols, **S. Phillips** and A. Soderlund. Resilience-based Optimization of Investigation Satellite Swarm via an Artificial Honeybee Colony Algorithm. In Proceedings of the 2022 AIAA SciTech Forum. National Harbor, MD 2023.
- [31] K. Miller, A. Soderlund and **S. Phillips**. Multi-agent Control of Chaser Satellites using Games with Lexicographic Preferences. In Proceedings of the 2022 AIAA SciTech Forum. National Harbor, MD 2023.
- [30] C. W. Hays, T. Henderson, A. Soderlund and **S. Phillips**. Enabling Proliferated Space Sensor Awareness Constellations Through Topological Observability. In Proceedings of the 2022 AIAA SciTech Forum. National Harbor, MD 2023.
- [29] A. Soderlund and **S. Phillips**. Robust Formation Control of a Distributed System with Sporadic Asynchronous Measurements. In Proceedings of the 2022 Conference on Decision and Control. Cancun, MEX. 2022.
- [28] C. Staudt, C. Zagaris and **S. Phillips**. Sliding Mode Control of Satellite Formation Proximity Operations under Imperfect Communications. In Proceedings of the 2022 AAS/AIAA Astrodynamics Specialist Conference. Charlotte, NC. 2022.
- [27] W. Flores, C. Zagaris and **S. Phillips**. A Function for Observability Analysis of Multi-agent Spacecraft Formation Trajectories. In Proceedings of the 2022 AAS/AIAA Astrodynamics Specialist Conference. Charlotte, NC 2022.
- [26] H. Lei, M. Shubert, K. Lang and **S. Phillips**. Deep Reinforcement Learning for Multi-Agent Autonomous Satellite Inspection. In Proceedings of the 44th Annual AAS Guidance, Navigation and Control (GN&C) Conference. Breckenridge, CO 2022.
- [25] A. Soderlund and **S. Phillips**. Autonomous Rendezvous and Proximity Operations of an Underactuated Spacecraft via Switching Controls. In Proceedings of the AIAA Scitech 2022 Forum. San Diego, CA 2022.
- [24] C. Petersen, **S. Phillips**, D. Hustig-Schultz, R. G. Sanfelice. Towards Hybrid Model Predictive Control for Computationally Aware Satellite Applications. In Proceedings of the Workshop on Computation-Aware Algorithmic Design for Cyber-Physical Systems, Virtual, May 2021
- [23] J. M. Brewer, P. Tsiotras, K. Lang, **S. Phillips**. Falsification-based Verification for Multi-Mode Spacecraft Attitude Control Systems. In Proceedings of the 2021 American Control Conference. New Orleans, LA 2021
- [22] F. Zegers, **S. Phillips**, and W. Dixon. Consensus over Clustered Networks with Asynchronous Inter-Cluster Communication. In Proceedings of the 2021 American Control Conference. New Orleans, LA 2021
- [21] C. Petersen, **S. Phillips**, K. Hobbs, and K. Lang. Challenge Problem: Assured Satellite Proximity Operations. In Proceedings of the AIAA/AAS Spaceflight Mechanics Meeting. Charlotte, NC 2021
- [20] **S. Phillips**, C. Petersen, and R. Fierro. Spacecraft Formation Control with Intermittent Measurement and Communication. AIAA Scitech 2021 Forum. Nashville, TN 2021
- [19] A. Soderlund, A Zaman, C. Petersen, and **S. Phillips**. Hybrid Satellite Rendezvous and Proximity Operations. AIAA Scitech 2021 Forum. Nashville, TN 2021

- [18] F. Parra and **S. Phillips**. A Case Study on Auction-Based Task Allocation Algorithms in Multi-Satellite Systems. AIAA Scitech 2021 Forum. Nashville, TN 2021
- [17] K. Lang, C. Klett, K. Hawkins, E. Feron, P. Tsiotras, and **S. Phillips**. Formal Verification Applied to Autonomous Spacecraft Attitude Control. AIAA Scitech 2021 Forum. Nashville, TN 2021
- [16] M. Mercier, **S. Phillips**, M. Shubert and W. Dong. Terrestrial Testing of Multi-Agent, Relative Guidance, Navigation, and Control Algorithms. 2020 IEEE/ION Position, Location and Navigation Symposium (PLANS), Portland, OR 2020.
- [15] **S. Phillips**, C. Petersen, R. Fierro. Resilient Spacecraft Formation Control Under Malfunctioning Communications. AIAA Scitech 2020 Forum. Orlando, FL, Jan 2020.
- [14] **S. Phillips**, and R. G. Sanfelice. Observer-based Synchronization of Multi-agent Systems using Intermittent Measurements. In *Proceedings of the Conference on Decision and Control*. Dec 2019. Nice France. (accepted)
- [13] D. Lavell, **S. Phillips**, and R. G. Sanfelice. A Hybrid PID Design for Asymptotic Stabilization with Intermittent Measurements. In *Proceedings of the Conference on Decision and Control*. pages 737-742. Dec 2018. Miami FL.
- [12] **S. Phillips**, R. S. Erwin, and R. G. Sanfelice. Robust Exponential Stability of an Intermittent Transmission State Estimation Protocol. In *Proceedings of the American Control Conference*. pages 622-627. June 2018. Milwaukee WI.
- [11] A. Duz, **S. Phillips**, A. Fagiolini, R. G. Sanfelice and F. Pasqualetti. Stealthy Attacks in Cloud-Connected Linear-Impulsive Systems. In *Proceedings of the American Control Conference*. pages 146-152. June 2018
- [10] **S. Phillips**, A. Duz, F. Pasqualetti and R. G. Sanfelice. Hybrid Attack Monitor Design to Detect Recurrent Attacks in a Class of Cyber-Physical Systems In *Proceedings of the 2017 IEEE 56th Annual Conference on Decision and Control*. pages 1368-1373. Dec 2017
- [9] **S. Phillips**, and R. G. Sanfelice. On Asymptotic Synchronization of Interconnected Hybrid Systems with Applications. In *Proceedings of the American Control Conference*. pages 2291-2296, May 2017
- [8] **S. Phillips**, Y. Li, and R. G. Sanfelice. A hybrid consensus protocol for pointwise exponential stability with intermittent information. In *Proceedings of the 10th IFAC Symposium on Nonlinear Control Systems*. pages 146–151, August 2016
- [7] Y. Li, **S. Phillips**, and R. G. Sanfelice. On Distributed Observers for Linear Time-invariant Systems Under Intermittent Information Constraints. In *Proceedings of the 10th IFAC Symposium on Nonlinear Control Systems*. pages 654–659, August 2016
- [6] **S. Phillips**, and R. G. Sanfelice. Robust Synchronization of Interconnected Linear Systems over Intermittent Communication Networks, In *Proceedings of the American Control Conference*, pages 5575–5580, June 2016.
- [5] **S. Phillips**, and R. G. Sanfelice. Robust Synchronization of Two Linear Systems over Intermittent Communication Networks, In *Proceedings of the Conference on Decision and Control*, pages 5569–5574, December 2015.
- [4] Y. Li, **S. Phillips**, and R. G. Sanfelice. Results on Incremental Stability for Hybrid Systems, In *Proceedings of the Conference on Decision and Control*, pages 3089–3094, December 2014.
- [3] **S. Phillips**, and R. G. Sanfelice. A framework for modeling and analysis of robust stability for spiking neurons, In *Proceedings of the American Control Conference*, pages 1414–1419, June 2014.
- [2] **S. Phillips**, and R. G. Sanfelice. Results on the asymptotic stability properties of desynchronization in impulse-coupled oscillators, In *Proceedings of the American Control Conference*, pages 3278–3283, June 2013.
- [1] **S. Phillips**, R. G. Sanfelice, R. S. Erwin. On the synchronization of two impulsive oscillators under communication constraints, In *Proceedings of the American Control Conference*, pages 2443–2448, July 2012.

Presentations

Invited Presentations.....

- [1] Data Sparsity in a Data Rich World, *2026 American Control Conference Pre-Workshop - Data Driven Methods in Satellite Resilience*, New Orleans LA, May 2026
- [2] Space Control Autonomous Systems, *AFRL Autonomy Community of Practice*, Virtual April 2026
- [3] Local INtelligent Network of Collaborative Satellites Laboratory: An Overview, *AFRL Discover Pre-Workshop, Space Trusted Autonomy*, Dayton OH, March 2026
- [4] Space Trusted Autonomy, *USA-SWE Technology Collaboration Meeting*, Virtual/Stockholm SWE, March 2026
- [5] Increasing Resilience, Capability and Capacity Through On-orbit Satellite Autonomy, *Nordic Space Defense Workshop*, Gausta NOR, Jan 2026
- [6] Local Intelligent Network of Collaborative Systems Lab: An Overview, *NRO/AFRL TIM*, Albuquerque NM, Mar 2025
- [7] Distributed Satellite Autonomy-Challenges, Approaches and Recent Results, *UNM ECE Graduate Student Seminar*, Albuquerque NM, Mar 2025
- [8] AFRL Space Trusted Autonomy, *USSF/SSC Integrated Technology Needs Meeting*, Los Angeles CA, Mar 2025
- [9] Challenges in Resilience in Distributed Satellites Systems (and some recent work), *SMC-IT Workshop on Satellite Resilience*, Mountain View CA, Jul 2024
- [10] Resilient Distributed Satellite Autonomy, *NRL Seminar Series*, Washington DC, Jan 2024
- [11] Benchmark Framework for Trusted and Safe Multi-Satellite Autonomy, *RV CS Innovation Portfolio Program*, Washington DC, Jan 2024
- [12] Cyber-Robust Control for Proliferated Multi-Satellite Systems, *RV CS Innovation Portfolio Program*, Washington DC, Jan 2024
- [13] Distributed Autonomy for Rapid Tasking and Multi-Perspective Targeting, *RV CS Innovation Portfolio Program*, Washington DC, Jan 2024
- [14] Assured Collaborative Satellite Autonomy, *2022 UNM Space and Aerial Robotics Workshop*, University of New Mexico, Jun 2022
- [15] On Resilient Distributed Autonomous Satellite Systems, *Seminar for NASA/NRO/AFRL Space Trusted Autonomy Working Group*, Virtual, Mar 2022
- [16] Trusted Distributed Intelligent Satellite Systems, *3TES, 1SOPS and 4SOPS*, Schriever SFB, Jan 2022
- [17] Collaborative Satellite Mission Autonomy, *AFRL/RVSW Program Technical Quality Review*, Virtual, Dec 2020
- [18] Collaborative Satellite Mission Autonomy, *AFRL/RVSW Program Technical Quality Review*, Virtual, Dec 2020
- [19] Collaborative Satellite Mission Autonomy, *AFRL-NRL Technical Interchange Meeting*, Virtual, Nov 2020
- [20] Collaborative Satellite Mission Autonomy, *AFRL Autonomy Scientific Advisory Broad Review*, Virtual, Oct 2020
- [21] Hybrid Systems and Analysis: Applications to Multiagent Systems, *RVSW Tech Talks*, Albuquerque, New Mexico, Apr 2020
- [22] Coordination and Control of Hybrid Systems over Intermittent Communications, *University of Arizona*, Tucson, Arizona, USA, Mar 2019

[23] Synchronization in Networked Hybrid Systems with Applications to Intermittent Communication, *Los Alamos National Laboratory* Los Alamos, New Mexico, USA, Nov 2017

[24] Lightning talk: Interconnection of Networks of Hybrid System with Applications to Attack Detection *UCOP LFRP Cybersecurity RFP Workshop UCSC-Silicon Valley*, California, USA, May 2017

Conference Presentations.....

[1] Next-Generation Autonomous Satellite Systems, *2023 AIAA Intelligent Systems Workshop* Boulder, CO 2023

[2] Spacecraft Formation Control with Intermittent Measurements and Communications, *2021 AIAA SciTech Forum*. Nashville, TN, Jan 2021

[3] Resilient Spacecraft Formation Control Under Malfunctioning Communications, *2020 AIAA SciTech Forum*. Orlando, FL, Jan 2020

[4] Observer-based Synchronization of Mutli-agent Systems using Intermittent Measurements, *Conference on Decision and Control*. Nice, FRA, Dec 2019

[5] Robust Exponential Stability of an Intermittent Transmission State Estimation Protocol, *American Control Conference*. Milwaukee, WI, June 2018

[6] Hybrid Attack Monitor Design to Detect Recurrent Attacks in a Class of Cyber-Physical Systems, *Conference on Decision and Control* Melbourne, Victoria, AUS, Dec 2017

[7] On Asymptotic Synchronization of Interconnected Hybrid Systems with Applications *American Control Conference* Seattle, Washington, USA, May 2017

[8] A hybrid consensus protocol for pointwise exponential stability with intermittent information *10th IFAC Symposium on Nonlinear Control Systems* Monterey, California, USA, August 2016

[9] Robust Synchronization of Interconnected Linear Systems over Intermittent Communication Networks *American Control Conference* Boston, Massachusetts, USA, July 2016

[10] Robust Synchronization of Two Linear Systems over Intermittent Communication Networks *Conference on Decision and Control* Osaka, Japan, Dec 2015

[11] A framework for modeling and analysis of robust stability for spiking neurons *American Control Conference* Portland, Oregon, USA, June 2014

[12] Results on the asymptotic stability properties of desynchronization in impulse-coupled oscillators *American Control Conference* Washington DC., USA, June 2013

[13] On the synchronization of two impulsive oscillators under communication constraints *American Control Conference* Montreal, Canada, July 2012

[14] Biological Trajectory Recording of Bee Paths and Analysis for Implementation to Micro Air Vehicles *2011 Arizona Space Grant Symposium* Tempe, Arizona, USA, April 2011

Poster Presentations.....

[1] Trusted Distributed Autonomy Demonstration Experiment: Simulation-Based Results *Small Sat Conference*, Logan, UT, 2023

[2] Catalog Maintenance of Close Proximity Satellite Systems on Closed Natural Motion Trajectories. *45th Rocky Mountain AAS GN&C Conference*, Breckenridge, CO, 2023

[3] Distributed Multi-Satellite Space Domain Awareness *Space Domain Awareness Conference*, Maui, HI, 2018

[4] Distributed Estimation and Consensus under Intermittent Communication for Space Situational Awareness *Air Force Research Lab Space Scholars Symposium*, Albuquerque, New Mexico, USA, August 2017

[5] Robust analysis and control of interconnected spiking neurons through hybrid systems modeling *Workshop on Brain Dynamics and Neurocontrol Engineering*, St. Louis, Missouri, USA, June 2017

- [6] Estimation and Synchronization of Multi-agent Systems Using Tools for Hybrid Systems *Baskin School of Engineering Open House*, Santa Cruz, California, USA, March 2017
- [7] Estimation and Synchronization of Multi-agent Systems Using Tools for Hybrid Systems *CITRIS @ UC Santa Cruz Open House*, Santa Cruz, California, USA, April 2016
- [8] Estimation and Synchronization of Multi-agent Systems Using Tools for Hybrid Systems *12th Annual Graduate Research Symposium*, Santa Cruz, California, USA, April 2016
- [9] Estimation and Synchronization of Multi-agent Systems Using Tools for Hybrid Systems *UCSC Research Review Day*, Santa Cruz, California, USA, October 2015
- [10] Estimation and Synchronization of Multi-agent Systems Using Tools for Hybrid Systems *CITRIS Day 2015*, Berkeley, California, USA, October 2015
- [11] On Desynchronization of Impulsive Oscillators for Coordination of Actions of Multiple Players *NSF Connection One Semi-Annual Meeting*, Scottsdale, Arizona, USA, May 2012
- [12] Optimal Cognitive Network Reconfiguration in Adversarial Conditions with Delays Using Game Theoretical Methods *Air Force Research Lab Space Scholar Symposium*, Albuquerque, New Mexico, USA, August 2012
- [13] On the Synchronization of Impulsive Oscillators For Decentralized Rendezvous *NSF Connection One Semi-Annual Meeting*, Scottsdale, Arizona, USA, May 2012
- [14] Simultaneous Synchronization Rendezvous (SSR) in Decentralized Cognitive Radio Networks *Air Force Research Lab Space Scholar Symposium*, Albuquerque, New Mexico, USA, August 2011
- [15] American Society of Mechanical Engineers Human Power Vehicle Competition: Speed Class *Engineering Senior Design Day*, Tucson, Arizona, USA, May 2011
- [16] Large Scale Stirling Engines *University of Arizona Honors College Undergraduate Research Symposium* Tucson, Arizona, USA, December 2010
- [17] Large Scale Stirling Engines *University of Arizona Alumni Student Showcase* Tucson, Arizona, USA, November 2010

Academic and Service Work

Air Force Research Laboratory Liaison to AME Program Board <i>Aerospace and Mechanical Engineering Program Board, Tucson, AZ</i>	University of Arizona 2019-present
Ph.D. Dissertation Committee Member <i>Student - Sarah Cleese, Gainesville, FL</i>	University of Florida 2026
Ph.D. Dissertation Committee Member <i>Student - Karthik Sivaramakrishnan, Albuquerque, NM</i>	University of New Mexico 2026
Committee Member <i>MIT-Lincoln Labs IRAD Review Board, Arlington, VA</i>	MIT-LL 2026
MS Thesis Proposal Committee Member <i>Student - Connor Staudt, Dayton, OH</i>	Air Force Institute of Technology 2025
Ph.D. Dissertation Proposal Committee Member <i>Student - Karthik Sivaramakrishnan, Albuquerque, NM</i>	University of New Mexico 2025
Ph.D. Dissertation Committee Member <i>Student - Spenser Hallyburton, Durham, NC</i>	Duke University 2025
Ph.D. Dissertation Committee Member <i>Student - Vignesh Sivaramakrishnan, Albuquerque, NM</i>	University of New Mexico 2024
Committee Member <i>MIT-Lincoln Labs IRAD Review Board, Arlington, VA</i>	MIT-LL 2024

Ph.D. Dissertation Committee Member <i>Student - Kristina Miller, Champaign, IL</i>	University of Illinois, Urbana-Champaign 2024
M.S. Thesis Committee Member <i>Student - Kealy Murphy, Dayton, OH</i>	Air Force Institute of Technology 2024
Ph.D. Dissertation Committee Member <i>Student - Christopher Hays, Daytona Beach, FL</i>	Embry-Riddle Aeronautical University 2023
Ph.D. Dissertation Proposal Committee Member <i>Student - Nicholas Yeilding, Wright-Patterson Air Force Base, OH</i>	Air Force Institute of Technology 2023
Ph.D. Dissertation Proposal Committee Member <i>Student - Christopher Hays, Daytona Beach, FL</i>	Embry-Riddle Aeronautical University 2023
Ph.D. Dissertation Proposal Committee Member <i>Student - Alex (Chandler) Nichols, Tuscaloosa, AL</i>	University of Alabama 2023
Committee Member <i>MIT-Lincoln Labs IRAD Review Board, Arlington, VA</i>	MIT-LL 2023
Ph.D. Dissertation Pre-Defense Committee Member <i>Student - Christopher Hays, Daytona Beach, FL</i>	Embry-Riddle Aeronautical University 2023
MS Thesis Proposal Committee Member <i>Student - Wilson Flores, Dayton, OH</i>	Air Force Institute of Technology 2022
Ph.D. Dissertation Proposal Committee Member <i>Student - John M. Brewer, Atlanta, GA</i>	Georgia Institute of Technology 2021
Ph.D. Dissertation Committee Member <i>Student - Mohammad Maadani, Tucson, AZ</i>	University of Arizona 2021
Steering Committee Member <i>Computer Engineering Graduate Student Society, Santa Cruz, CA</i>	UC Santa Cruz 2015-2017
Graduate Student Representative <i>Aerospace and Mechanical Eng. Dept. Machine Shop Committee, Tucson, AZ</i>	University of Arizona 2011-2014
Graduate Student Advisor <i>Student Chapter of American Society of Mechanical Engineering, Tucson, AZ</i>	University of Arizona 2011-2013
Undergraduate Student Representative <i>Aerospace and Mechanical Eng. Dept. Machine Shop Committee, Tucson, AZ</i>	University of Arizona 2009-2011
Founding Treasurer <i>Student Chapter of American Society of Mechanical Engineering, Tucson, AZ</i>	University of Arizona 2009-2011
Steering Committee <i>Student Chapter of American Society of Mechanical Engineering, Tucson, AZ</i>	University of Arizona 2009

Professional Activities

- **Associate Editor** of the IEEE Transactions on Control Systems Technologies, 2024 - present.
- **Associate Editor** of the IEEE Transactions on Aerospace and Electronic Systems, 2023 - present.
- **Invited Committee Member** of Best Student Paper Award at the AACC/IEEE American Control Conference in New Orleans, LA 2027.
- **National Chair** of the Autonomous GNC Program of the American Astronomical Society Rocky Mountain Section Guidance Navigation and Control Conference, Breckenridge, CO 2027
- **Organization Committee Member** of the Intelligent Systems Technical Committee Workshop in Dayton,

OH 2026.

- **Committee Member** of Best Student Paper Award at the AACC/IEEE American Control Conference in New Orleans, LA 2026.
- **Session Organizer and Chair** of Sessions "Spacecraft Control on Complex Manifolds" and "Safety Verification and Optimization Protocols for Spacecraft" at the AACC/IEEE American Control Conference in New Orleans LA, 2026
- **Session Organizer and Chair** of Sessions "Space Domain Awareness and Sensing" and "Emerging Threats and Challenges in National Security" at the AIAA ASCEND Conference in Washington DC, 2026
- **Co-Organizer** of the DoD "5th AI in Contested Space Workshop" in Dayton, OH 2026
- **Technical Discipline Lead - National Security Topic** of the AIAA ASCEND Conference in Washington DC, 2026
- **Session Organizer** of Sessions "Space Domain Awareness" and "Emerging Threats and Challenges in National Security" at the AIAA ASCEND Conference in Washington DC, 2026
- **Session Chair** of Sessions "Trusted Space Autonomy I" and "Trusted Space Autonomy II" at the AIAA SciTech Forum Conference in Orlando, FL 2026
- **Co-Organizer** of the DoD "4th AI in Contested Space Workshop" in Dayton, OH 2025
- **Technical Discipline Lead - Space Security and Protection** of the AIAA ASCEND Conference in Las Vegas, NV 2025
- **Session Organizer** of Sessions "Orbital Debris Management" and "Sensing in Space, Detection, Characterization and Mitigation" at the AIAA ASCEND Conference in Las Vegas, NV 2025
- **Session Chair** of Sessions "Orbital Debris Management," "Sensing in Space, Detection, Characterization and Mitigation," and "Space Sector Growth: Connections and Inclusion" at the AIAA ASCEND Conference in Las Vegas, NV 2025
- **Organizer and Chair** of Invited Sessions "Autonomous Spacecraft Decision-Making" and "Safe and Constrained Spacecraft Control" in the 2025 AACC/IEEE American Control Conference, Denver, CO 2025.
- **Associate Editor** of the Conference Editorial Board for AACC/IEEE American Control Conference, 2019 - 2024.
- **Program Committee** of the "28th ACM International Conference on Hybrid Systems: Computation and Control", Irvine, CA, 2025
- **Organizer and Session Chair** of Invited Sessions "Trusted Satellite Autonomy I" and "Trusted Satellite Autonomy II" in the 2024 AIAA SciTech Forum, Orlando, FL 2024
- **Associate Editor** of the Conference Editorial Board for IEEE Conference on Decision and Control, 2019 - 2024.
- **Member and Line of Effort Lead** of the NATO SCI-SAS-351 Panel on Allied Space Deterrence Framework 2021 - present
- **Technical Discipline Lead - Space Security and Protection** of the AIAA ASCEND Conference in Las Vegas, NV 2024
- **Co-Organizer** of the DoD "3rd AI in Contested Space Workshop" in Dayton, OH 2024

- **Co-Organizer** of the DoD “2nd AI in Contested Space Workshop” in Colorado Springs, CO 2023
- **Co-Organizer** of US DoD and Swiss Defense “Swarm Intelligence Workshop” in Zurich CHE 2023
- **Organizer** of Invited Sessions “Satellite Autonomy I” and “Satellite Autonomy II” in the 2024 AIAA SciTech Forum, Orlando, FL 2024
- **Organizer and Chair** of Invited Sessions “Safe Spacecraft Control” and “Autonomous Satellite Control Systems” in the 2023 AACC/IEEE American Control Conference, San Diego, CA 2023.
- **Organizer and Chair** of Tutorial Session “Safe and Constrained Rendezvous, Proximity Operations and Docking” in the 2023 AACC/IEEE American Control Conference, San Diego, CA 2023.
- **DoD/AFRL Liason** to the Aerospace and Mechanical Engineering Department Program Advisory Board at the University of Arizona, 2019 - present.
- **Chair** of sessions “Agent Based Systems” at the IEEE Conference on Decision and Control in Cancun, MEX, 2022.
- **Chair** of sessions “Autonomous Satellite Control Systems”, “Satellite Attitude Control” and “Safe Spacecraft Control” at the AACC/IEEE American Control Conference in Atlanta, GA, USA, 2022.
- **Chair and Co-chair** of sessions “Autonomous Aerospace Systems - I”, “Networked Control”, “Autonomous Aerospace Systems - II” at the IFAC Networked & Autonomous Air & Space Systems (NAASS) in Santa Fe, NM, USA.
- **Member of** IEEE Control Systems Society Hybrid Systems Technical Committee from 2016 - present.
- **Co-chair** of session “Hybrid and Switched Systems” at the 2012 American Control Conference in Montreal, QC, CA.
- **Reviewer** of technical papers submitted to the journals: IEEE Transactions on Automatic Control; IEEE Transactions on Signal Processing; IEEE Transactions on Control Systems Technology; and Automatica.
- **Reviewer** of technical papers submitted to the conferences: IEEE Conference on Decision and Control; IEEE American Control Conference; IEEE Conference on Control Technologies; AIAA Scitech; European Control Conference.

Mentorship

Summer Faculty Fellowship Program.....

2025: Prof. Chris Petersen	University of Florida
2025: Ms. Sarah Clees (student)	University of Florida
2022: Prof. David Copp	University of California, Irvine
2022: Prof. Romiel Sandhu	Stoney Brook University
2022: Dr. Daniel Grange (student)	Stoney Brook University
2019: Prof. Wei Dong	University of Texas, Rio Grande Valley

National Research Council Fellows.....

2026 - Present: Dr. Dan Calderon	Senior Research Fellow, University of New Mexico
2024 - Present: Dr. Vignesh Sivaramakrishnan	Postdoctoral Fellow, Ph.D. from University of New Mexico
2025: Prof. Allie Haymen	Senior Research Fellow, University of Colorado, Boulder, CO
2023: Prof. Neera Jain	Senior Research Fellow, Purdue University, IN

2020 - 2022: Dr. Alexander Soderlund

Postdoctoral Fellow, Ph.D. from The Ohio State University

Air Force Research Lab Interns.....

2026: David Peyton (COSMIAC)	<i>B.S. from University of New Mexico, NM</i>
2026: Gabrielle Newcomb (ML-RCP)	<i>B.S. from University of Arizona, AZ</i>
2026: Edward Jung (Space Scholar)	<i>Ph.D. from University of Texas Austin, TX</i>
2025-2026: Cade Byers (Phillips Scholar)	<i>B.S. from University of Kentucky, KY</i>
2025-2026: Sarah Clees (Space Scholar)	<i>Ph.D. from University of Florida, FL</i>
2025-2026: Brayden Forrester (COSMIAC)	<i>B.S. from University of New Mexico, NM</i>
2025-2026: Oscar Jed Chuy (Space Scholar)	<i>Ph.D. from Georgia Institute of Technology, GA</i>
2025-2026: David Flores (COSMIAC)	<i>B.S. from University of New Mexico, NM</i>
2025: Alex Maynes-Pointon (COSMIAC)	<i>B.S. from University of New Mexico, NM</i>
2025: Cade Byer (Phillips Scholar)	<i>B.S. from University of Kentucky, KY</i>
2025: Alexis Limary (Phillips Scholar)	<i>H.S. from Albuquerque Academy, NM</i>
2024: Bennet Outland (SMART)	<i>B.S. from South Dakota School of Mines, SD</i>
2024: Kole Barkman (PCIP)	<i>B.S. from University of Kansas, KS</i>
2024: Seth Kercheff (NSIN)	<i>B.S. from Purdue University, IN</i>
2023-2025: Kelsey Knowlson (COSMIAC)	<i>M.S. & B.S from University of New Mexico, NM</i>
2023-2024: Tristan Thomas (Space Scholar & NSF INTERN)	<i>Ph.D. in progress from Clemson University, SC</i>
2023: Jacob Jackson (Phillips Scholar)	<i>B.S. from Virginia Institute of Technology, VA</i>
2023: Thomas Chung (Phillips Scholar)	<i>Student at La Cueva High School, NM; now at Yale University, NJ</i>
2023: Sabrina Hwang (Phillips Scholar)	<i>Student at University of Washington, WA; now at Verus Research supporting AFRL</i>
2021-2024: Kristina Miller (Space Scholar & SMART)	<i>Ph.D. at University of Illinois, Urbana-Champaign, IL; now Civilian at AFRL</i>
2021-2024: Chandler Nichols (Space Scholar & SMART)	<i>Ph.D. at University of Alabama, Tuscaloosa, AL; now at Aerospace Corp.</i>
2023: Daniel Grange (Space Scholar)	<i>Ph.D. at Stoney Brook University, NY; now at Memorial Sloan-Kettering Institute</i>
2022: John Pendergrast (Space Scholar)	<i>B.S. at Massachusetts Institute of Technology, MA; now 2LT at AFIT</i>
2022: Gabriel Behredt (Space Scholar)	<i>Ph.D. at Georgia Institute of Technology, GA; now at Aerospace Corp.</i>
2022: James Hand (Space Scholar)	<i>Ph.D. at Embry-Riddle University, FL</i>
2021-2022: Mario Sumali (Phillips Scholar)	<i>B.S. at Stanford University, CA now at NRO</i>
2020: Federico Zegers (Space Scholar)	<i>Ph.D. at University of Florida, FL now at JHU-APL</i>
2020: Dawn Hustig-Schultz (Space Scholar & NSF INTERN)	<i>Ph.D. at University of California, Santa Cruz, CA now at Supernal</i>
2019-2021: Caleb Bisig (Space Scholar)	<i>M.S. at the University of Cincinnati, OH</i>
2020-2023: Anonto Zaman (Phillips Scholar & NDSEG)	<i>B.S. at Georgia Institute of Technology, GA; M.S. at Stanford University, CA now at Verus Research supporting AFRL</i>
2019-2020: Alexander Soderlund (Space Scholar)	<i>Ph.D. at The Ohio State University, OH now Civilian at AFRL</i>
2019: Fernando Parra (COSMIAC)	<i>M.S. & B.S. at University of New Mexico, NM now at COSMIAC</i>

2019: Devon Alcorn (Space Scholar)

*M.S. at Virginia Institute of Technology, VA
now at Northrop-Grumman
B.S. at University of Hawaii, HI
now at Remitly*

2019: Simon Huang (Phillips Scholar)

Outreach and Volunteer Activities

Volunteer: Operational Member <i>Albuquerque Mountain Rescue Council</i>	Albuquerque, NM <i>Feb 2022-present</i>
Volunteer: Orientation to Cave Rescue <i>National Cave Rescue Commission</i>	Grants, NM <i>Jan 2025</i>
Volunteer: Operational Core Curriculum Application Panel <i>Albuquerque Mountain Rescue Council</i>	Albuquerque, NM <i>Jan 2025</i>
Volunteer: Operational Core Curriculum Application Panel <i>Albuquerque Mountain Rescue Council</i>	Albuquerque, NM <i>Jan 2024</i>
Volunteer: Member-at-Large (Board of Directors Position) <i>Albuquerque Mountain Rescue Council</i>	Albuquerque, NM <i>Nov 2022- Nov 2023</i>
Volunteer: CODE <i>Lobo Rainforest, public coding education event</i>	Albuquerque, NM <i>Oct 2018</i>
Panelist: Graduate Student Researcher Panel <i>CMPE 200: Research and Teaching in Computer Science and Engineering</i>	UC Santa Cruz, CA <i>Oct 2017</i>
Volunteer: Trail Building at Teaching Rock <i>Pinnacles Climber Appreciation Day, Pinnacles National Park, CA</i>	Pinnacles National Park <i>Oct 2017</i>
Volunteer: Adopt-a-crag Glen Canyon <i>Bay Area Climbers Coalition, San Francisco, CA</i>	Glen Canyon State Park <i>Apr 2017</i>
Judge: Field Competition <i>VEX Robotics State Championship</i>	Santa Clara, CA <i>Feb 2017</i>
Lecturer: Hybrid System Lab: Intro. to Feedback Control <i>Girls in Engineering</i>	UC Santa Cruz <i>July 2016</i>
Volunteer: Adopt-a-crag Goat Rock <i>Bay Area Climbers Coalition, Saratoga, CA</i>	Castle Rock State Park <i>Apr 2016</i>
Volunteer: MESA Achievement Day at UC, Santa Cruz <i>Mathematics, Engineering, Science Achievement</i>	UC Santa Cruz <i>Mar 2016</i>
Volunteer: Adopt-a-crag Summit Rock <i>Bay Area Climbers Coalition, Saratoga, CA</i>	Castle Rock State Park <i>Oct 2015</i>
Volunteer: Adopt-a-crag Indian Rock <i>Bay Area Climbers Coalition, Saratoga, CA</i>	Castle Rock State Park <i>Apr 2015</i>
Lecturer: Control Engineering for High School Teachers <i>Mathematics, Engineering, Science Achievement</i>	University of Arizona <i>Aug 2011</i>
Mentor: High School Students for Summer Internship <i>Hybrid Dynamics and Controls Lab</i>	University of Arizona <i>June 2010</i>
Lecturer: Intro. to Vicon Camera System and Feedback Control <i>Hybrid Dynamics and Controls Lab</i>	University of Arizona <i>Oct 2009</i>

News Articles

[1] Shield AI flies Hivemind autonomy software on satellite in orbit, Space News, 2026

- [2] AFRL Early Career Award, AFRL Press Release, 2024
- [3] Tools for Enabling Autonomous Multi-satellite Systems, Biz Journals, 2023
- [4] Jack Baskin and Peggy Downes-Baskin Fellowship Announcement, UC Santa Cruz News, 2017